

ANISH RAY

*PhD Student in Mathematics, University of Houston
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PERSONAL DETAILS

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EDUCATION

Ph.D. in Mathematics **August 2025 – Present**
University of Houston, USA
Current GPA: 3.778

M2 Mathématiques et Applications, Spécialité Analyse, Arithmétique, Géométrie **2024 – 2025**
Institut de mathématique d'Orsay, Université Paris-Saclay, France
Overall grade: 12.01 (*Assez Bien*)

M.Sc. in Mathematics **2020 – 2023**
Universität Münster, Germany
Overall grade: 1.8 (*Gut*)

B.Sc. (Hons.) in Mathematics and Computing **2017 – 2020**
Institute of Mathematics and Applications, Bhubaneswar (IMA), affiliated with Utkal University, Bhubaneswar, Odisha, India
Grade: First Class with Distinction

RESEARCH INTERESTS

Analytic Number Theory, L -functions, Spectral Theory of Automorphic Forms, Automorphic Representations, Arithmetic Statistics, Equidistribution Problems, and Homogeneous Dynamics.

PREPRINTS

1. *On the Constants of the Lang-Trotter Conjecture for CM Elliptic Curves.*
arXiv:2309.09938 [math.NT]

RESEARCH AND READING PROJECTS

PIMS – First Year Interest Groups **October 2025 – May 2026**
Group Reading Project on Primes and the Theory of the Riemann Zeta Function, Pacific Institute for the Mathematical Sciences
Supervisor: Dr. Nicol Leong, Postdoctoral Fellow, University of Lethbridge

M2 Thesis **December 2024 – June 2025**
On Subconvexity of L -functions on $U(n) \times U(n+1)$ in the Depth Aspect
Supervisor: Prof. Farrell Brumley, IMJ-PRG, Sorbonne Université, France Grade: 14.5/20 (*Bien*)

Master's Thesis**September 2022 – March 2023***On the Constants of the Lang-Trotter Conjecture for CM Elliptic Curves*

Supervisor: Prof. Dr. Christopher Deninger, Mathematisches Institut, Universität Münster, Germany Grade: 1.0 (Highest Grade)

Research Project**April 2023 – April 2024***Asymptotics of the Number of Solutions of Forms Having Prime Coordinates via the Hardy-Littlewood Circle Method*

Supervisor: Prof. Julia Brandes, Chalmers University of Technology, Gothenburg, Sweden

Research Project**November 2022 – October 2023***Developed a New Large Sieve Inequality by Introducing Suitable Weight Functions*

Supervisor: Prof. Dr. Jan-Christoph Schlage-Puchta, Universität Rostock, Germany

Bachelor's Project**February 2020 – May 2020***Some Problems on Elementary Number Theory*

Supervisor: Dr. Sudhansu Sekhar Rout (currently Asst. Prof., NIT Calicut, Kerala, India)

Summer Project (Visiting Summer Research Programme)**June 2019 – July 2019***Reading Project on the Spectral Theory of Unbounded Operators*

Supervisor: Dr. Shyam Ghoshal, TIFR Centre for Applicable Mathematics (CAM), Bengaluru, India

ACADEMIC AWARDS

AMS–NSF ICM 2026 Graduate Student Travel Grant**2026***Awarded by the American Mathematical Society supporting travel for poster presentation at ICM 2026 in Philadelphia.***Presidential Fellowship****2025 – 2027***Awarded by the University of Houston to recruit doctoral students with outstanding credentials. The fellowship provides \$2,000 in addition to the Graduate Assistantship for the first two years of the PhD programme.***Graduate Tuition Fellowship****2025 – Present***Competitive fellowship awarded by the University of Houston Graduate School covering in-state tuition and mandatory fees for PhD students.***M2 Internship Allowance****2025***Awarded by the Institut de Mathématiques de Jussieu-Paris Rive Gauche (IMJ-PRG) to support thesis completion.*

Monthly allowance of €660 for three months.

IDEX Excellence Scholarship**2024 – 2025***Awarded by the Université Paris-Saclay to only 80 international students pursuing a master's degree.*

Total of €10,000 for ten months, plus €850 in travel expenses.

Graduation Scholarship**2021 – 2022***Awarded by the International Office, WWU Münster.*

Total of €1,500 for three months.

Merit Scholarship**2015***One-time merit scholarship of INR 5,000 awarded by the Kendriya Vidyalaya Sangathan (KVS) for securing 10 CGPA in AISSE 2015.*

CONFERENCE AND SEMINAR TALKS

International Congress of Mathematicians**2026***Philadelphia*

Poster presentation: “A Computational Comparison of Lang-Trotter and Hardy-Littlewood Constants for CM Elliptic Curves” at the ICM in July 2026.

Graduate Student Paper Presentation <i>Organized by the UH-AMS Graduate Chapter, University of Houston</i>	2026
Probabilistic Number Theory: Law of Large Numbers and CLT in Number Theory <i>Universität Münster</i>	2022
Linear Algebraic Groups: Connected Solvable Groups and Maximal Tori <i>Universität Münster</i>	2022
Elliptic Curves, Tate Module and the Weil Pairing <i>Universität Münster</i>	2022
Class Field Theory <i>Universität Münster</i>	2021
Zeta and L-functions in Algebraic Number Theory <i>Universität Münster</i>	2021

CONFERENCES, SUMMER SCHOOLS, AND WORKSHOPS ATTENDED

Recent Perspectives on Moments of L-functions <i>University of Texas at Dallas</i>	2026
Workshop on Automorphic Forms and Number Theory <i>IMJ-PRG, Sorbonne University, Paris</i>	2025
Summer School on Inclusive Paths in Explicit Number Theory <i>Banff International Research Station (BIRS) at the University of British Columbia – Okanagan, Canada</i>	2023
Summer School on Automorphic Forms <i>Alfréd Rényi Institute of Mathematics, Budapest</i>	2022

TEACHING

Teaching Assistant, University of Houston <i>MATH 2415 – Calculus III</i>	Fall 2026
Teaching Assistant, University of Houston <i>MATH 2451 – Accelerated Calculus II</i>	Spring 2026
Teaching Assistant, University of Houston <i>MATH 2305 – Discrete Mathematics</i>	Fall 2025, Spring 2026
Associate Faculty, Cheenta Ganit Kendra (https://cheenta.com/) <i>Online tutoring platform for aspirants of Mathematics Olympiads and national-level entrance examinations.</i> Conducted lectures, problem-solving sessions, and created and graded assignments, quizzes, and examinations.	2017 – 2018

ACADEMIC SERVICE AND OUTREACH

Undergraduate Mentor, Directed Reading Program (DRP) <i>Department of Mathematics, University of Houston</i> Mentoring an undergraduate student in a reading project on Selberg's elementary proof of the Prime Number Theorem.	Fall 2026
Proofreader, University of Houston Mathematics Contest <i>The University of Houston hosts a major annual Mathematics Contest for high school and middle school students, featuring individual and team examinations across multiple tracks.</i> https://mathcontest.uh.edu/	2026

Head of Technical Affairs and Problem Selection Committee

2018

Scholastic Aptitude Test in Mathematics (SATM), conducted by the student association of the Institute of Mathematics and Applications, Bhubaneswar, as part of its annual techno-cultural fest. The event focuses on spreading awareness about higher education in mathematics and organises training camps for selected high school students in Odisha, India.

<https://sites.google.com/view/satm-ima>

SKILLS

<i>Programming</i>	C, C++, PYTHON, JAVASCRIPT, JAVA, VISUAL BASIC
<i>Typesetting</i>	L ^A T _E X
<i>Markup</i>	HTML
<i>Stylesheet</i>	CSS
<i>Office Software</i>	MS WORD, MS EXCEL, MS POWERPOINT, MS ACCESS
<i>Languages</i>	Bengali (mother tongue) English (professional) German (proficient, A2) French (proficient, A1) Hindi (native) Sanskrit (intermediate)